

Daniel Guo

Mathematics
(Computing + ML)
Student @UWaterloo

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Links:

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Relevant Skills:

Languages:	Frameworks:
• SQL • C/C++ • Typescript • Javascript • Bash • HTML/CSS • Python • Java • C#	• Vite/React • NextJS • FastAPI • PostgreSQL • NodeJS • Tensorflow • OpenCV • ReportLab • LangGraph
Tools:	Misc Concepts:
• Git/Github • Linux • Docker • MS Office • GraphQL	• REST APIs • Websocket API • Data Structures & Algorithms • Neural Networks • Computer Vision

Relevant Experiences

- AI Backend Development Intern**
ZigZag Technologies Inc. May 2026 – Aug 2026
- Developer on *Outpost*, a **fully autonomous, agent-run** retail platform (**hardware control + CV + payments + distributed systems**)
 - Built **AI agent harness** and tools including **CV derived store analytics**, and automated local **market trend analysis** with **time series data** using the **AI-native framework Jingway**
 - Working across **hardware interfaces, vision pipelines**, and **agent orchestration** in a **closed-loop physical environment**

- Automations & Machine Learning Intern**
Legends Haul June 2026 – May 2026
- Automated manual AP and AR data entry processes for shipping logistics, **saving up to 4 hours of work per day**, and rectifying **over 600 past data entry errors** utilizing **OCR based image detection** with dynamic Excel report cross-checking
 - Implemented two stage signature detection algorithm at **97% accuracy**:
 - **OpenCV contour-based heuristics** + finetuned CPU optimized **YOLO V8-based model with hard negative mining training loop** to distinguish noisy signatures vs handwritten notes on forms at **96% accuracy** (from 82% in just five runs)

- Controller Software Developer | Stat Team Lead**
First Robotics Competition Team 8082 Sept 2020 – May 2022
- **Programmed robot control systems** with **Java WPILib**, integrating motor controllers and sensor feedback loops
 - **Managed statistics and strategy analysis** to optimize team performance during competitions, **earning 2nd place rank**

Interesting Projects (Cont. Next Page)

- VR 3D Modelling App | C#, Meta SDK, Unity**
- Developed with team for hackathon at Carnegie Mellon, won **Honorable Mention** (out of 300 Submissions)
 - Integrated **VR Controller Vector Tracking** in 3D – Allowing for easy creation of 3D art models

Notable Awards:

1st Place (Out of ~100)

University of Waterloo Stock Valuation Pitch Competition

Honorable Mention (Out of 874)

Carnegie Mellon University Nexhacks Hackathon Competition

Presidential Scholar's Distinction

University of Waterloo

Certificate of Distinction Award

Canadian Senior Math Competition

Silver (Vancouver Regionals)

First Robotics Competitions

2nd Place (Out of ~300)

Simon Fraser World Schools Debate

School Champion

University of Waterloo FGH Mathematics Competitions

Leadership:

School District Board Rep (out of 2):

Earl Marriott Secondary

Head Debate Coach:

White Rock Christian Academy

Student Council Treasurer:

Earl Marriott Secondary

Rotary Club President:

Earl Marriott Secondary

Math and English Tutor:

Novabridge Learning

Math & CS Club Co-Lead:

White Rock Christian Academy

Interesting Projects (Cont.)

ASCII Fluid Dynamics | *HTML Canvas, Vanilla JS*

- Implemented Jos Stam's 1999 Stable Fluids paper's fluid simulation for my personal website danielguo.net
- Tracked density and velocity via **semi-lagrangian scheme** (prevents lag/buildup), with **Helmholtz-Hodge projection**
- **Dynamically injected** fitting ASCII (based on velocity and density) into **HTML Canvas**

Spotifixer Chrome Extension | *JS, Manifest V3*

- Chrome extension with **4.9 stars** and over **250 downloads** (no marketing, pure SEO)
- Detects missing Spotify lyric states, and **dynamically fetches** and **injects** the lyrics retrieved from open source, **entirely client side**
- Implemented **robust song-matching logic**, handling variations in song titles and **fall back** for sources

Sketch/Calligraphy Alignment Algorithm | *Python OpenCV*

- Designed **image-comparison pipeline** to identify and annotate deviations in hand-drawn sketches for art practice.
- Accounts for scale and angle differences by aligning the images via **template matching** (using **canny edge detection**)
- Then finds "matching" points via **shear transforms** guided by **ORB matching**

Tensorflow Chess Bot | *Python, Tensorflow, Pandas*

- Extracts board features for games via CNN and dense layers, predicts **probability gradients** for the "next move".
- Achieved **73% accuracy** in high elo move prediction through supervised reinforcement learning.

Poetry Recommendation Algorithm | *Python, Tensorflow*

- Used **Hugging Face Sentence Transformer Library** to embed poetry dataset items into **FAISS vector database**
- Serves initials poems via **random sampling** across **k-means clusters** to determine initial interests
- Generates new poems by **aggregating preference vectors** then ranking candidates by **cosine similarity**